

Permutations



1

a 120

e 1

b 42

f 11

c 120

g 210

d 120

h 20 160

2 362 880

3 120

4 120

Probabilities with permutations

5 $\frac{1}{2}$

6

a 24

b 6

c $\frac{1}{4}$

7

a $\frac{5}{6}$

b 72

8

a $\frac{3}{28}$

b $\frac{15}{28}$

9

a 120

b $\frac{1}{5}$

c $\frac{1}{120}$

10

a 40 320

b $\frac{1}{40\,320}$

11

$$\begin{array}{l} \text{a} \quad \frac{1}{3} \\ \text{e} \quad \frac{4}{5} \end{array}$$

$$\text{b} \quad \frac{2}{3}$$



$$\text{c} \quad 0$$

$$\text{d} \quad \frac{1}{5}$$

12

$$\begin{array}{l} \text{a} \quad \frac{1}{120} \\ \text{e} \quad \frac{3}{5} \end{array}$$

$$\begin{array}{l} \text{b} \quad \frac{1}{120} \\ \text{f} \quad \frac{3}{5} \end{array}$$

$$\text{c} \quad \frac{1}{120}$$

$$\text{d} \quad \frac{2}{5}$$

Applications

$$13 \quad 120$$

$$14 \quad 5040$$

15

$$\text{a} \quad \frac{1}{120}$$

$$\text{b} \quad \frac{1}{120}$$

$$\text{c} \quad \frac{1}{120}$$

$$\text{d} \quad \frac{1}{5}$$

16

$$\text{a} \quad 17\,576\,000$$

$$\text{b} \quad \frac{1}{17\,576\,000}$$

$$\text{c} \quad \frac{1}{13}$$

17

$$\text{a} \quad 720$$

$$\text{b} \quad 120$$

$$\text{c} \quad \frac{1}{6}$$

18

$$\text{a} \quad \frac{1}{90}$$

$$\text{b} \quad \frac{1}{90}$$

$$\text{c} \quad \frac{1}{45}$$

$$\text{d} \quad \frac{4}{9}$$

19

$$\frac{1}{2\,793\,510\,720}$$

$$20 \quad \frac{7}{11}$$

21

$$\text{a} \quad 1$$

$$\text{h} \quad 29\,889$$



22 a $\frac{1}{210}$

b $\frac{1}{35}$

23 a $\frac{1}{156}$

b $\frac{1}{1716}$

24 $\frac{1}{720}$

25 $\frac{1}{22}$